

The Vector-Constraint Index

A deterministic test of whether a problem is actually being solved

Andrew Delgado, Creator 1 LLC · September 29, 2026 · Spec CC BY 4.0, code MIT · DOI 10.5281/zenodo.23039677

“Institutions will try to preserve the problem to which they are the solution.”

Clay Shirky. Named the Shirky Principle by Kevin Kelly, 2010.

The Shirky Principle tells you problem preservation exists. The Vector-Constraint Index tells you where.

DOCTRINE

1. **VCI sets no standard.** The problem owner's desired outcome is the standard. VCI tests whether a solver is actually delivering it.
2. **The owner is whoever bears the cost of the problem,** not whoever funds the solver. Where the two differ, the owner's outcome wins.
3. **Purpose: certify a genuine attempt.** Catching handwaving is the byproduct. An honest solver should pass with nothing to fear.
4. **Function, not label.** Calling an effort a study, pilot, partnership, or case study changes which rules apply to it, never whether VCI applies.
5. **Burden of proof is on the solver.** Every question is answered Yes, No, or Can't show it. Can't show it counts as No.
6. **Verdicts describe structure, not intent.** A failed gate means the claim is not supported, not that anyone is lying (Hanlon's Razor). The structural finding is provable from documents. Intent is not.
7. **Deterministic.** The same answers always produce the same verdict. Thresholds and question text are published and versioned.
8. **VCI applies to itself and to the evaluator.** The evaluator states their own stake before starting. Section 9 runs VCI on VCI.

Status. Release candidate. Freezes as v1.0 after an independent blind validation run. See the invitation to falsify, next page.

This specification is published as Release Candidate 3 (RC3).

The core mechanics (the Delgado Vector Law, the closure test, the six vector roles, and kill forks K0 to K6) have been run against five cases with known outcomes: smallpox, the 1955 malaria eradication programme, Guinea worm, polio, and the brown tree snake on Guam. In that run the framework broke one of its own assumptions: the brown tree snake's entry routes had not been counted. The fix is now part of the rules. That run was done by the author with outcomes already known. It is not the blind test that freezes v1.0.

No diagnostic tool is finished until it meets the adversarial world. This specification is published under CC BY 4.0, and the evaluator under the MIT License, so that anyone can try to break it. A gap found is the framework getting stronger.

1. **Submit an edge case.** If you have a proposal, public program, or contract whose lane the closure test cannot resolve, send the public documents.
2. **Report a wrong verdict.** If an effort that genuinely ended is marked Problem preserved, or an open-ended effort passes the kill forks unchecked, send the case and the evidence.
3. **Meet the standard for change.** Any change to the core rules must name the documented case that exposed the failure and state the rule that fixes it. It must also rerun every prior benchmark case and report any verdict that changes, with the reason.

Every accepted break is credited by name in the changelog, unless the finder asks to stay anonymous. The first entry is the author's own: the brown tree snake correction in RC2.

Where to send it: open an issue on the project repository, or write to andrew.delgado@crea8or1.com with "VCI break" in the subject.

The framework does not ask for trust. It asks to be tested.

1 Definitions

If problem and solution stay undefined, a preserver redefines them to fit what they are already doing. Every term here is testable.

Problem

A measurable gap between the current state and the state the owner wants, which costs the owner something and persists without intervention.

ELEMENT	TEST
Owner	Someone bears the cost. Where millions do, the measured harm (incidents, losses, exposures) stands in for them. A body claiming to speak for owners counts only if it does not also fund or run the solver.
Current state	Measured.
Desired state	Measurable, and set by the owner.
Cost	What the gap takes from the owner: money, safety, time, health, trust.
Changeability	Some intervention could, in principle, close it.

Missing any element means there is no defined problem yet, and nothing can be claimed as solving it. If no intervention could change the gap, it is a **condition** to plan around, not a problem to fund. The test cuts both ways: it stops fixes being sold for the unfixable, and it catches a solvable problem relabeled as a condition so nobody expects it to end.

Supporting terms

Symptom	The observable effect: outages, bites, breaches, cases.
Agent	The thing that causes the symptom: the snake, the virus, the attacker, the leak.
Vector	A distinct route by which the agent reaches, moves through, or re-establishes itself in the owner's space. Defined in full below.
Constraint	What a set of vectors has to pass through. A Terminal solution interdicts it.
Solver	Whoever claims to be fixing the problem: contractor, agency, researcher, clinician, charity, vendor, program, or yourself.
Resources	Whatever the solver asks the owner to keep giving: money, time, votes, trust, attention, authority, patience.
Evaluator	Whoever runs VCI. States their own stake before starting.

Vector

A distinct route by which the problem's agent reaches, moves through, or re-establishes itself in the owner's space.

Closure test (how N is counted). Two routes are the same vector if one interdiction closes both. They are different vectors if closing one leaves the other open. N is the smallest number of independent closures that would shut every route. Routes that share a constraint collapse into one, so the count cannot be padded. A route left open after a closure is by definition its own vector, so the count cannot be shrunk.

Every run answers all six roles in writing, even when the answer is "none." Leaving a role blank is how routes get missed. The worked column is the case that forced this definition.

ROLE	QUESTION	BROWN TREE SNAKE ON GUAM
Source	Where does the agent come from, and can it be eliminated?	Native range in New Guinea and nearby islands. Permanent.
Entry	How does it get inside the boundary?	Sea cargo, air cargo and aircraft, vessels. Each port or airfield is a door.
Movement	How does it move once inside?	Surface contact. One constraint, covering canopy, ground, culverts, and the waterline.
Regeneration	How does it replace itself inside?	Eggs in refuges. This is why a full hatch cycle is the clearance clock.
Induced	What routes does the intervention itself create?	Displacement into neighboring ground, answered by directing it toward capture.
Adaptive	Can it invent new routes?	No record of it.

A source is not a vector; it is what feeds them. Whether the source can be eliminated decides whether the doors can ever close. How an agent moves and how it gets in are counted separately: a constraint on movement does not close a door.

Counted vector	Backed by a record: interceptions, detections, genetic tracing, incident data.
Suspected vector	Listed separately with the test that would confirm it, and not counted in N until confirmed. This stops a solver inventing vectors to justify open-ended work, and stops anyone waving off a real one.
Not vectors	Symptoms (outages, bird losses). The agent itself. Conditions of will: no funding, nobody cares. Neglect is a finding against whoever should be acting, never a vector or a lane.

Solution

An intervention that moves the measured state to the owner's desired state, is shown to be the cause of that change, holds under real conditions, and ends the solver's necessary involvement apart from defined maintenance.

TEST	WHERE VCI CHECKS IT
Reaches the owner's outcome	K2, K6
Proven to be the cause	Gate 6, reversal
Holds under the conditions that caused the failure	Gate 5, validation
Has an end	K5

2 Claim types

Most handwaving lives here: selling one of these as another. Every type is legitimate when named honestly. At intake the solver declares a type, and the type sets which tests apply. Only the disguise fails.

CLAIM TYPE	WHAT IT DOES	ITS HONEST END
Solution	Closes the gap between the current state and the owner's desired state.	The owner's outcome is reached and the solver's necessary involvement ends, apart from defined maintenance.
Partial solution	Closes a stated, measured part of the gap.	The stated fraction is reached.
Mitigation	Reduces what the gap costs the owner. The gap stays open.	A stated cost reduction is reached, then a dated reassessment.
Management	Holds an ongoing problem steady: an Adaptive problem, or the doors of an Inflow problem.	A metric plus a dated sunset review where the work is cut if the metric has not moved.
Research	Answers a question about the gap.	The answer is delivered, then handed to a solver or the approach is declared unable to solve it.
Pilot	Tries an approach at small scale to inform a decision.	A decision date: scale it, change it, or kill it.

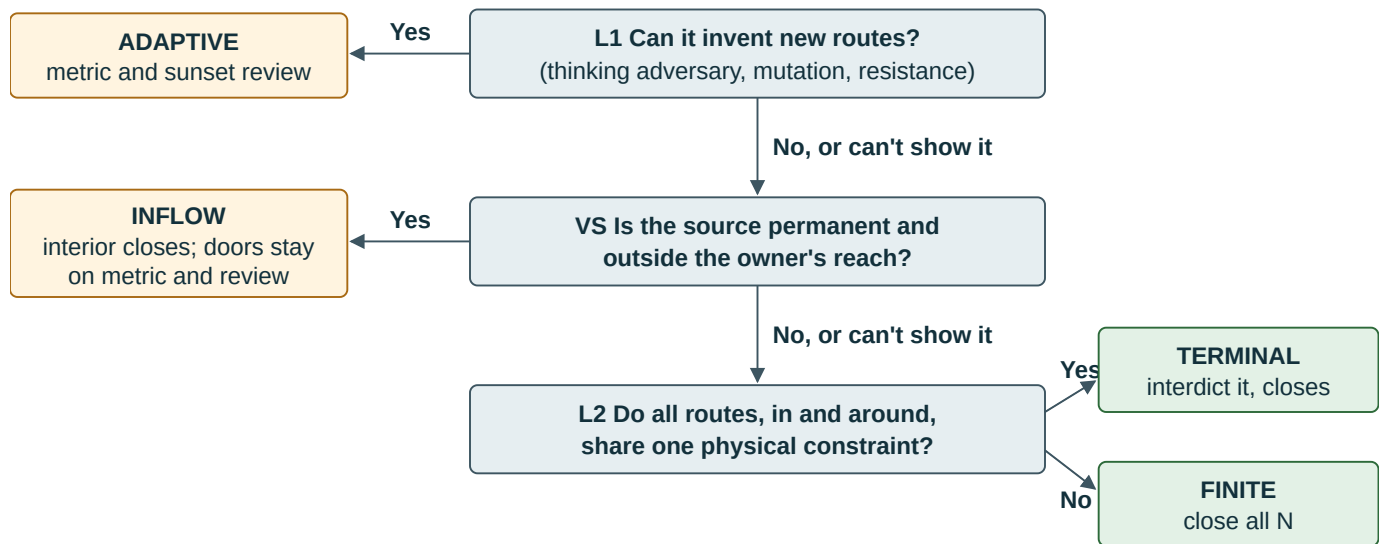
Mislabeled claim (stop). A mitigation or management effort presented as a solution; research presented as a fix; management sold on a Terminal or Finite problem, or on interior ground of an Inflow problem; a solution claimed against a whole Adaptive or Inflow problem without naming the sub-problem it closes; a pilot renewed with no decision.

Maintenance rule. Recurring upkeep of a deployed intervention is allowed in any lane when it runs at a fixed interval, at a fixed price, and is measured against a published decay or wear curve. Open-ended advisory, study, or monitoring is not maintenance.

3 Lanes: the Delgado Vector Law

Whether a problem is allowed to be ongoing depends on three facts: whether its source can be eliminated, how many vectors it has, and whether they share a constraint. This is the part of VCI that separates a legitimately open-ended effort from a solvable problem kept open because it pays. It extends Goldratt's Theory of Constraints (1984) from system throughput to whether a problem can be closed at all.

A problem closes when its source can be eliminated and its routes, both in and around, can be closed: at one shared constraint (Terminal) or one by one (Finite). It is legitimately ongoing only where a permanent source keeps pressing on its doors (Inflow) or where it invents new routes (Adaptive), and even then only under a metric and a sunset review. Some problems have no end. None has no answer.



TERMINAL

The source can be eliminated, and every entry and movement route passes through one shared constraint. Interdict the constraint and the problem closes. Terminal means an end exists, not that it is cheap or fast.

FINITE

The source can be eliminated, and the routes are a fixed, countable set with no shared constraint. The problem closes when all N are closed. Progress is counted X of N.

INFLOW

The agent comes from a permanent source outside the owner's reach, through a finite set of doors. The interior closes like a Terminal or Finite problem. The doors stay ongoing, under a metric and a sunset review tied to the source.

ADAPTIVE

The problem can invent new routes on its own (a thinking adversary, mutation, resistance). Ongoing work is legitimate here, but only with a metric and a sunset review.

Decomposition rule

Large problems fail K2 until they are split, which is correct behavior. Split the problem into sub-problems, assign each its own lane, and judge every claim at the sub-problem level. An Adaptive parent can hold Terminal children. Cybersecurity is Adaptive, but SQL injection as a class closes with parameterized queries, and password theft as a vector closes with hardware keys. Selling ongoing management on a Terminal child is a Mislabeled claim. An Inflow parent splits the same way: the brown tree snake on Guam is an Inflow problem whose interior is a Terminal child, one parcel at a time, and whose doors are the ongoing part. A permanent sentinel is earned at the doors and nowhere else.

Research lane

Research is legitimate, and it is not open-ended either: research ends when the question is answered, whichever way it comes out. When the claim type is Research or Pilot, the kill forks test different things.

FORK	SOLUTION TYPES	RESEARCH AND PILOT
K2	Owner's outcome, measurable, agreed	The question, plus the decision it will feed
K3	Failure condition written before the work	The 'no' result and analysis plan written before data; all results reported (file drawer, Rosenthal, 1979)
K4	Failure hit, kept going	Answer came back 'no' and was reframed as promising
K5	End state	End date and handoff. Pilot: decision date to scale, change, or kill. Same question asked again, or wider, is perpetual study
K6	Measured on the owner's outcome	Answers the original question, not a swapped one

"Further research needed" is allowed once, and only with a narrower question. A pilot renewed without a decision is rerouted and judged as the claim type it has become. A case study that only documents is not a claim; the moment it asks for continued resources tied to the problem, K0 catches it.

4 How to run it

Answer rule

Every question takes **Yes**, **No**, or **Can't show it**, which counts as No. A few questions allow **Does not apply**, only under the stated condition. Each question carries an evidence line saying what counts as Yes.

Stage

Before resources are committed. Escape hatches are open. A solver who fixes a gap now proceeds clean.

After resources are committed. Escape hatches are closed for what was already spent. A stop verdict stands for that spend. The solver may resubmit the next tranche as a fresh claim, with this result on record.

A surgeon does not get to define malpractice after the patient is on the table. The hospital can still demand a proper plan before the next operation.

Newcomer rule

A solver with no prior offers is flagged **No record** and skips the track record section without penalty. Otherwise VCI locks out every new solver and protects incumbents, the opposite of its purpose.

Evaluator stake

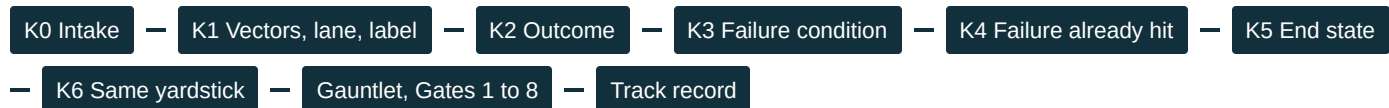
Before starting, the evaluator writes down any stake they hold in the outcome. VCI applies to whoever runs it.

Scoring

Kill forks stop the run the moment one cannot be passed. Survivors run the gauntlet:

0 soft fails	GENUINE ATTEMPT
1 to 4	CREDIBLE, WITH GAPS
5 or more	UNPROVEN

Run order



Most unsupported claims stop within six questions, at K3, K4, or K5. Every stop names what it caught, so the verdict carries a citation instead of an opinion.

5 Kill forks

Some forks already tell the whole story. If the solver cannot get past one, the other questions do not matter. Each fork ends the run with a named verdict.

K0 Intake

Shirky Principle (Shirky; named by Kelly, 2010)

Are resources being spent or requested while someone bears the cost of the problem?

Plain: Is anyone paying, in money, time, trust, or harm, while this effort goes on?

Counts as Yes: Yes when there is a budget, contract, grant, subscription, program, or ongoing ask of any kind. The label on the effort (study, pilot, case study, partnership) does not matter.

No: **OUTSIDE VCI SCOPE** Yes: solver declares a claim type, evaluator states their stake, continue.

K1 Vectors, lane, and label

Delgado Vector Law

V0 Are all six vector roles listed: source, entry, movement, regeneration, induced, and adaptive? **soft**

VC Was the vector count N set with the closure test, with each counted vector backed by a record? **soft**

L1 Can the problem invent new routes on its own, through a thinking adversary, mutation, or resistance?

Yes only with evidence of new routes appearing over time. A claim that it might adapt, with no record of it, counts as No. A route the intervention itself creates is an Induced vector and must be counted, but it does not make the problem Adaptive.

VS Does the agent come from a permanent source outside the owner's reach, shown by evidence?

Yes when the source is shown to be permanent and out of reach: a native population in its home range, another jurisdiction, a natural reservoir. A source is not permanent because nobody has funded removing it. Lack of money or will is a neglect finding, never a vector or a lane.

L2 Do all routes, entry and movement alike, pass through one shared physical constraint?

M1 Does what the solver will actually deliver match the claim type they named?

M2 (Solution or partial solution in the Adaptive or Inflow lane) Does the solution name the specific sub-problem it closes completely?

M3 (Management in the Inflow lane) Is the ongoing work limited to the doors, with the interior handled as a problem that closes?

Management in a Terminal or Finite lane, or No on M1, M2, or M3: **MISLABELED CLAIM** Escape before commitment: relabel honestly and continue.

K2 Desired outcome

Hypothetico-deductive method; Heilmeier Catechism

Is the owner's desired outcome stated as a number or clear observable, and has the owner agreed to it?

- **Mitigation:** Is the cost reduction the owner wants stated as a number or clear observable, and has the owner agreed to it?
- **Research:** Is the question stated, together with the decision its answer will feed?
- **Pilot:** Is the decision the pilot will inform stated (scale, change, or kill), with the measurement that will decide it?

Plain: What exactly counts as fixed, and who agreed to that?

Evidence: The owner is whoever bears the cost of the problem, not necessarily whoever funds the solver. Where millions bear the cost, the measured harm (incidents, losses, exposures) stands in for them. A body that claims to speak for owners counts only if it does not also fund or run the solver.

Fail: **NOT A TESTABLE CLAIM** Escape before commitment: State the outcome now as a number or observable and have the owner sign off.

K3 Failure condition

Popper, 1934/1959; pre-registration (Ioannidis, 2005)

Was a result that would prove the approach wrong written down before work or resources began?

- **Research:** Were the result that means 'no' and the analysis plan written down before data was collected?

Plain: Before starting, did they say what result means they were wrong?

Evidence: Yes only with a dated record (quote, contract, proposal, ticket, protocol, registry) that predates the work.

Fail: **UNFALSIFIABLE** Escape before commitment: State the failure condition now and agree to be held to it.

K3R Full reporting Research, Pilot only

Rosenthal, 1979 (file drawer problem)

Are all results, positive and negative, reported to the owner?

Plain: Do they share the bad results too?

Evidence: No when only favorable results are published or reported.

Fail: **UNFALSIFIABLE** No escape hatch.

K4 Failure already hit

Lakatos, 1970 (degenerating research programme)

Is the effort clear of a failure it already hit? Its stated failure result has not happened, and no prior effort by this solver on this problem, using this approach, hit its failure condition and carried on.

- **Research:** Is the effort clear of a reframed 'no'? The answer has not already come back 'no' and then been presented as promising.

Plain: Did they already hit their own wall and keep going with new explanations?

Evidence: No when the failure condition was met and the effort continued with new explanations, new 'hidden issues', or a changed environment that was not predicted in advance.

Fail: **PROBLEM PRESERVED** No escape hatch. A genuinely new approach can re-enter at K2 as a fresh claim, with this failure on its record.

K5 End state

Shirky Principle; NASA NPR 7120.5 key decision points

Is there a measurable condition where the solver's necessary involvement ends, apart from defined maintenance?

- **Solution:** Is there a measurable condition where the solver's necessary involvement ends, apart from defined maintenance?
- **Partial solution:** Is there a measurable condition, the stated fraction of the gap closed, where this work ends?
- **Mitigation:** Is there a stated cost reduction and a dated reassessment?
- **Management:** Is there a metric for the ongoing work and a dated sunset review where it is cut if the metric has not moved? In the Inflow lane, the review also checks whether the source still exists.
- **Research:** Is there an end date and a named handoff: to a solver, or a declaration that this approach cannot solve it?
- **Pilot:** Is there a decision date to scale, change, or kill it?

Plain: When does this end?

Evidence: Defined maintenance (fixed interval, fixed price, measured against a published decay or wear curve) does not count against an end state. Open-ended advisory, study, or monitoring does.

Fail: **PROBLEM PRESERVED** Escape before commitment: State the end condition now.

K5R Perpetual study Research only

Lakatos, 1970; Platt, 1964

Is this a new question, or a narrower one than any already answered? (Not the same question asked again, and not a wider one.)

Plain: Are they studying something already studied, just again?

Evidence: 'Further research needed' is allowed once, and only with a narrower question.

Fail: **PROBLEM PRESERVED** No escape hatch.

Is success measured on the owner's outcome from K2, not on a metric the solver introduced later?

- **Research:** Does the work answer the original question, not a new one swapped in?

Plain: Are they grading themselves on the thing you asked for?

Evidence: No when the reported success is activity (hours, scans, meetings, reports produced) or a proxy the solver picked, instead of the owner's outcome.

Fail: **MOVED THE GOALPOSTS** No escape hatch.

6 The gauntlet

Claims that survive the kill forks answer every remaining question. Each No is a soft fail and comes with the follow-up question to put to the solver. Question 7.4 is the one fatal item: two consecutive phases that eliminate nothing.

ID	QUESTION	BASIS
GATE 1 PROBLEM DEFINITION		
1.1	Was the owner's outcome measured before work began, to set a baseline? <i>Plain: What did it look like before they touched it?</i> Yes means: A dated measurement of the K2 outcome.	Hypothetico-deductive method
1.2	Are two or more competing causes named? <i>Plain: What else could be causing it?</i> Yes means: At least two alternatives to the favored cause, in writing. Skipped for Research	Chamberlin, 1890 (multiple working hypotheses)
1.3	Was each competing cause ruled out by a test? <i>Plain: How did they rule the others out?</i> Yes means: A test or measurement per alternative. 'Experience' does not count. Skipped for Research	Platt, 1964 (strong inference)
1.4	Is the current approach stated, with why it falls short? <i>Plain: How is it done today, and what is wrong with that?</i> Yes means: A named current method and its measured limit.	Heilmeier Catechism
1.5	Is it stated what is new compared with past attempts? <i>Plain: What is different this time?</i> Yes means: A specific difference, not 'better' or 'innovative'.	Heilmeier Catechism
1.6	Is the payoff stated: who benefits, and by how much, if it works? <i>Plain: Who cares, and what changes for them?</i> Yes means: A named beneficiary and a measurable difference.	Heilmeier Catechism
GATE 2 FALSIFICATION		
2.1	Does the solver state what they lose if the failure condition is met? <i>Plain: What does it cost them to be wrong?</i> Yes means: A refund, unpaid labor, a performance hold-back, a public result, or similar.	Taleb, 2018 (skin in the game)
GATE 3 BOUNDARY		
3.1	Is the governing variable or constraint named? <i>Plain: What is the one thing driving the failure?</i> Yes means: One variable, or a tight group with a reason.	Derived from dimensional reduction (Buckingham, 1914)
3.2	Are exclusions stated: what the effort deliberately leaves out, and why? <i>Plain: What are they not touching?</i> Yes means: A written list of what is out of scope. An untreated route or refuge inside the boundary must appear here with a reason, or it is a neglect finding.	Derived from dimensional reduction (Buckingham, 1914)
3.3	Is it defined where the fix physically or logically stops? <i>Plain: Where does their fix end?</i> Yes means: A named part, circuit, area, process, or population.	Derived from dimensional reduction (Buckingham, 1914)
3.4	Is the scope unchanged since the start, or is each change tied to a named finding? <i>Plain: Has the job grown, and if so, why?</i> Yes means: No when scope grew without a finding that required it.	Derived from dimensional reduction (Buckingham, 1914)

GATE 4 VERIFICATION

- 4.1** Is a spec, drawing, protocol, or standard named that the work is built to? IEEE Std 1012 (verification)
Plain: What were they building it to?
Yes means: A named document. 'Industry best practice' does not count.
- 4.2** Does measured data show the spec was met? IEEE Std 1012 (verification)
Plain: Show the numbers, not a signed checklist.
Yes means: Test logs, measurements, inspection records.
Skipped for Research

GATE 5 VALIDATION

- 5.1** Was it tested under the same conditions that caused the failure? IEEE Std 1012 (validation)
Plain: Did they test it where it actually breaks?
Yes means: Same load, weather, traffic, population, or use. Bench-only counts as No.
Does not apply only if no field results exist yet. Bench results test Gate 4, not Gate 5 or 6.
- 5.2** Has it run clean for longer than the old interval between failures? IEEE Std 1012 (validation)
Plain: Has it held longer than it used to last?
Yes means: Clean time measured against the historical failure interval.
Does not apply only if no field results exist yet. Bench results test Gate 4, not Gate 5 or 6.
Skipped for Research

GATE 6 CONTROL

- 6.1** Does the failure return with the fix removed and stop with it restored, or is a safe equivalent shown? ABAB reversal design (Barlow and Hersen, 1984); Mill, 1843
Plain: Take it out, does the problem come back? Put it back, does it stop?
Yes means: A reversal test, a control group or area, or a documented reason reversal is unsafe plus the next-best comparison.
Does not apply only if no field results exist yet. Bench results test Gate 4, not Gate 5 or 6.
- 6.2** Are other changes made at the same time accounted for? Mill, 1843 (method of difference)
Plain: Did anything else change that could explain it?
Yes means: A list of concurrent changes and why each is ruled out.
Does not apply only if no field results exist yet. Bench results test Gate 4, not Gate 5 or 6.

GATE 7 PROGRESS

- 7.1** Did the last phase eliminate at least one candidate cause or option? Bayesian confirmation; Platt, 1964
Plain: What is ruled out now that was not before?
Yes means: Named candidates removed. The count of eliminated candidates must rise every phase.
Does not apply to a single-phase effort.
- 7.2** Did the last round produce a specific finding that changed the plan? Bayesian confirmation
Plain: What did the last round of spending teach them?
Yes means: A named finding. 'We gathered more data' does not count.
Does not apply to a single-phase effort.
- 7.3** Is the next phase narrower than the last? Bayesian confirmation
Plain: Is each step smaller and more targeted?
Yes means: A smaller scope, fewer candidates, or a tighter question.
Does not apply to a single-phase effort.

7.4	Has the effort avoided two consecutive phases that eliminated nothing? <i>Plain: Two rounds in a row with nothing ruled out is a fishing trip.</i> Yes means: No when two phases in a row ended with no candidate eliminated while uncertainty held or widened. A real discovery collapses old candidates, so honest work passes this. Fatal: Problem preserved Does not apply with fewer than two completed phases.	Lakatos, 1970; Bayesian confirmation
GATE 8 END STATE		
8.1	Is there a date or milestone for reaching the end state? <i>Plain: When?</i> Yes means: A calendar date or a dated milestone.	Heilmeier Catechism; NASA NPR 7120.5
8.2	Is the total cost to the end state stated? <i>Plain: How much, all in?</i> Yes means: A total, not a monthly rate with no end.	Heilmeier Catechism
8.3	Are closures counted, X of N vectors closed (or, at Inflow doors, X of N doors covered)? <i>Plain: How many routes are shut, out of how many?</i> Yes means: A count against the counted vector list from VC. Finite, Inflow lane only	Delgado Vector Law
8.4	If there is recurring maintenance, is it fixed interval, fixed price, and measured against a published decay or wear curve? <i>Plain: Is the upkeep scheduled and priced, or open-ended?</i> Yes means: No when recurring work is open-ended advisory, study, or monitoring. Does not apply when there is no recurring maintenance.	VCI maintenance rule
TRACK RECORD		
T0	Does the solver have prior offers on this problem or similar ones? A solver with no prior offers is flagged 'No record' and is not penalized. Track record questions are skipped for them.	Reference class forecasting (Kahneman and Tversky, 1979; Flyvbjerg, 2006)
T1	Are the prior offers listed? <i>Plain: What did they offer before?</i> Yes means: Offer, date, and owner for each.	Reference class forecasting (Flyvbjerg, 2006)
T2	For each prior offer, is the promised result shown next to the measured result? <i>Plain: What did they promise, and what actually happened?</i> Yes means: Promised versus delivered, side by side.	Reference class forecasting (Flyvbjerg, 2006)
T3	Did most prior offers deliver what they promised? <i>Plain: Do they usually deliver?</i> Yes means: More than half delivered the promised result.	Reference class forecasting (Kahneman and Tversky, 1979)
T4	Did prior work finish within the promised cost and time? <i>Plain: Were they on budget and on time?</i> Yes means: Actual versus promised cost and schedule.	Reference class forecasting (Flyvbjerg, 2006)
T5	Did any prior engagement reach its end state and actually end? <i>Plain: Have they ever finished a job and left?</i> Yes means: At least one closed engagement.	Shirky Principle

7 Across every arena

The gates never depended on a vendor. They test any claim that a problem is being solved, whoever makes it and whatever they ask for.

ARENA	K3 ASKED THERE	K5 ASKED THERE
Government program	What result would show this program failed?	When does the program sunset if it works?
Research grant	What finding would kill the hypothesis?	What result ends the study, and who gets the answer?
Medical treatment	What outcome means this is not working?	When do we stop or switch?
Nonprofit or charity	What number shows the problem is not shrinking?	What does "problem solved, we close" look like?
Software or subscription	What would prove the tool is not solving it?	When is it fixed so I can cancel?
Public safety	What figure would show the approach failed?	What level ends the surge?
Contractor or repair	If this part is swapped and the fault stays, what then?	When is the job done and the invoice final?
Your own project	What test result would prove my design wrong?	What is the ship condition?

The three-question short form

1. What result would prove your work wrong, and did you write it down before starting?
2. Can you show the failure with your fix out, and gone with it in?
3. What is the measurable condition where your involvement ends?

Anyone who is actually trying to solve the problem answers all three without strain. A problem that can invent new vectors answers the third with a metric and a sunset review.

8 Validation against history

A test that has not been tested is a claim. These cases have known outcomes. VCI was run on each as the claim stood at launch, and the verdict was compared with what happened.

Limitation, stated up front. This run was done by the author with outcomes already known. It checks that VCI routes real cases consistently. It is not the blind test. The blind test (Section 9) is what freezes v1.0.

Smallpox WHO Intensified Eradication Programme, 1967 to 1980		CONSISTENT
Lane	Terminal. Humans were the only host; no animal reservoir, no latent or persistent human infection. One host is one constraint.	
Claim	Solution. Outcome: zero cases worldwide. Ring vaccination narrowed each phase to cases and contacts (Gate 7).	
VCI verdict	Genuine attempt.	
Outcome	Last natural case in Somalia, 1977. Eradication declared at the 33rd World Health Assembly, May 8, 1980. Still the only human disease eradicated.	

Malaria WHO Global Malaria Eradication Programme, 1955 to 1969		CONSISTENT
Lane	Adaptive. Mosquito resistance to DDT was already emerging at launch and was used as the argument for launching quickly. Chloroquine resistance was confirmed in 1960.	
Claim	Solution: time-limited global eradication, largely through indoor residual spraying.	
VCI verdict	Mislabeled claim at K1 (M2). A solution claimed against a whole Adaptive problem without naming the sub-problems it closes.	
Outcome	Discontinued in 1969; eradication abandoned in favor of control. It did eliminate malaria in many temperate regions, consistent with the decomposition rule: Terminal children closed, the Adaptive parent did not.	

Guinea worm Guinea Worm Eradication Program, 1986 to present		CONSISTENT
Lane	At launch, Terminal: infection came through drinking water carrying infected water fleas. In 2012 infections in dogs were confirmed, which changed the vector list. Re-run: Finite (human and animal hosts, drinking water, undercooked aquatic animals).	
Claim	Solution. Outcome: zero human and animal infections, certified country by country.	
VCI verdict	Genuine attempt, end not yet reached. When the dog finding widened the problem, the program answered with targeted studies of dog behavior and water use and a new diagnostic for dogs, which narrows (Gate 7). This case tests the false-alarm guard: long duration alone must not read as preservation.	
Outcome	Human cases fell from an estimated 3.5 million (1986) to 10 provisional cases in 2025. 200 countries certified free. Animal infections rose slightly in 2025, driven by Cameroon and Angola, while Chad cut animal infections 47%.	

Polio Global Polio Eradication Initiative, 1988 to present CONSISTENT	
Lane	Finite: three wild poliovirus types, human hosts only. The oral vaccine can revert and circulate (cVDPV), an intervention that created its own vector, which VCI now requires to be counted (changelog item 10).
Claim	Solution, with closures countable X of N.
VCI verdict	Credible, with gaps. Passes the kill forks and counts closures. Fails T4: the 2000 target was missed, and later deadlines in 2012 and 2015 were missed.
Outcome	Wild types 2 and 3 certified eradicated in 2015 and 2019: 2 of 3 closed. Type 1 remains endemic in Afghanistan and Pakistan. 143 cVDPV cases were reported in 2025 as of September 17. Current targets: type 1 certification by 2027, cVDPV2 elimination by 2029.

Brown tree snake Guam, Habitat Management Unit 2011 to present, and island-wide RECLASSIFIED	
First pass	Filed as Terminal on movement alone: every move is a surface contact. That pass counted how the snake moves and never counted how it gets in. Corrected in RC2 (changelog item 13).
Lane	Inflow. Source: a native population in New Guinea and nearby islands, permanent. Entry: ports, airfield, vessels, a finite set of doors. Movement: one constraint, surface contact, including culverts and the waterline. Interior parcels are Terminal children; the doors are the ongoing part.
Claims	The HMU: partial solution on a Terminal child, a 55 ha block ringed by 3.6 km of snake barrier fence, with aerial acetaminophen bait. Cargo and port interdiction: management at the doors, legitimate in this lane with a metric and a review.
VCI verdict	HMU passes as a partial solution with a measurable end, a snake-free unit. Interdiction passes as door management. An island-wide effort needs an interior end state; ongoing management sold on interior ground would be Mislabeled (M3).
Outcome	Managers reported in 2023 that snakes inside the HMU were increasingly hard to find. The Navy reports the last live snake to escape Guam was in 2006. Barriers have been built; any claim that none exist is false and should not be repeated.

CASE	VCI VERDICT	ACTUAL	AGREEMENT
Smallpox	Genuine attempt	Eradicated 1980	Yes
Malaria GMEP	Mislabeled claim	Abandoned 1969	Yes
Guinea worm	Genuine attempt, open	10 human cases, 2025	Yes
Polio	Credible, with gaps	2 of 3 closed, dates missed	Yes
Guam, Inflow	HMU passes; doors ongoing	Snakes declining in unit	Yes, island-wide open

Negative controls are thin: only the malaria programme is a known failure in this set. The blind run needs more known failures so VCI's accuracy is shown in both directions.

9 Self-audit: VCI run on VCI

The first principle is that you must not fool yourself, and you are the easiest person to fool (Feynman, 1974). A tool that demands falsification from everyone else and has none itself gets thrown out on sight.

Claim type	Solution: a test that identifies genuine attempts.
Lane	Finite. The routes a problem-solving claim can dodge are enumerated in the kill forks and gauntlet. New dodges are added through the changelog.
K2 outcome	Two evaluators who do not know each other, given the same evidence, reach the same verdict, and verdicts predict how efforts turn out.
K3 failure conditions	VCI is wrong, and gets revised, if any of these occur: (1) efforts known to have succeeded are stopped at a kill fork; (2) efforts known to have failed or been preserved pass as Genuine attempts; (3) independent evaluators given the same evidence disagree on the verdict in more than 1 case in 10.
K5 end state	v1.0 freezes after one blind run: at least 20 historical cases with known outcomes, at least a third of them known failures, scored by at least two evaluators who have not seen the outcomes. After that, changes happen only through numbered versions.
K6 yardstick	Measured on evaluator agreement and verdict-outcome agreement, the K2 outcome.
Evaluator stake	The author holds IP he pitches with VCI scorecards. That is why the blind run uses outside evaluators.
Current verdict	CREDIBLE, WITH GAPS Fails 5.1 (not yet tested blind, under the conditions where it would fail) and 6.1 (no control comparison yet). Both close with the blind run.

10 Known objections

Every attack found so far, with its answer. A critic who finds their objection already answered stops attacking the tool and starts answering the questions.

Who made you a standard?

Nobody, and VCI claims no standard. The problem owner's outcome is the standard. An owner who wants VCI to govern their own resources adopts it with the clause in Section 11.

This is just the Heilmeier Catechism.

VCI builds on it and credits it. Heilmeier asks a solver to describe the plan. VCI tests whether the plan can be proven wrong, whether the fix is shown to be the cause, and whether the problem is allowed to be ongoing at all.

Real problems are complicated.

Complexity is allowed. The decomposition rule splits the problem and judges each part. Every part still ends at an outcome, an answer, a decision, or a sunset review.

Some problems never end.

Then they are Adaptive and ongoing work is legitimate, under a metric and a sunset review. What fails is a problem with an end being sold as one without.

Fixing it revealed hidden issues.

That happens honestly. It passes when each new issue comes with its own evidence (3.4) and its own failure condition. It fails when a stated failure condition was hit and the effort carried on (K4).

We are a study, a pilot, a case study.

Function, not label. Research ends at an answer, a pilot at a decision, and a case study is not a claim until it asks for continued resources.

Terminal means you expect it done by Friday.

Terminal means an end exists. It says nothing about cost or speed. Smallpox was Terminal and took thirteen years of the intensified programme.

Your own product needs maintenance.

Fixed interval, fixed price, measured against a published decay curve is maintenance and is allowed in every lane. Open-ended advisory is not.

This will kill new ideas.

The newcomer rule removes track record penalties, the Research lane gives early work its own honest end, and every escape hatch is open before commitment.

Escape hatches are a get-out-of-jail card.

They close once resources are committed. The verdict on what was spent stands. The hatch only lets the next tranche start as a fresh, pre-registered claim.

Honest discovery can widen uncertainty.

It can, and it still eliminates something. The measure is candidates eliminated, which must rise every phase. Two consecutive phases eliminating nothing is the fatal item 7.4.

The funder decides what success is.

The owner is whoever bears the cost. Where the funder is also the preserver, the owner's outcome wins. Where owners are diffuse, the measured harm stands in for them.

This is an accusation.

Verdicts describe structure, not intent. 'No exit condition and revenue tied to the problem staying open' is provable from documents. Motive is not claimed.

The evaluator has an agenda.

The evaluator states their stake before starting. VCI applies to whoever runs it, including its author (Section 9).

Metrics can be gamed.

K6 ties success to the owner's outcome from K2. A metric the solver chose later does not count.

You called the brown tree snake single-vector.

The first pass counted how it moves and not how it gets in. Counting both, with the closure test, puts it in the Inflow lane. The correction came from the framework's own flagship case and is logged in the changelog, which is how VCI is supposed to behave when it is wrong.

Nobody will fund fixing the culverts, so it is ongoing.

Neglect is not a lane. An untreated route inside the boundary is still closable. It is reported as a neglect finding against whoever should act, never used to justify open-ended work.

Your thresholds are arbitrary.

They are published, sit in one config file, and change only through a numbered version with the objection that caused the change.

11 Adoption clause

VCI claims no authority. An owner putting up their own resources can give it theirs. This is where directive language belongs.

[Owner] evaluates every proposal, renewal, and phase request under the Vector-Constraint Index, version **1.0-RC3** (C1-VCI-SPEC-2026.1).

Proposals must answer kill forks K0 through K6 in writing before resources are committed. A proposal that stops at a kill fork is returned without award. A renewal or next phase is evaluated after commitment, and a stop verdict on resources already spent is recorded.

Our desired outcome for this effort is:

Signed for the owner: _____ Date: _____

Promise ledger sheet

Log every offer when it is made, and log the result when it lands. Over time this builds the record that T1 to T5 draw from, and it makes pre-registration public instead of relying on the solver's own paperwork.

Offer	Date made	K2 outcome promised	K3 failure condition	K5 end state	Cost and time promised	Result delivered	Cost and time actual	Ended?

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Related tools

Sagan's baloney detection kit (*The Demon-Haunted World*, 1995) tests claims about the world; Bergstrom and West (*Calling Bullshit*, 2020) test data claims; Frankfurt (*On Bullshit*, 1986) defines the indifference to truth VCI detects in outcomes; Brandolini's law explains why a ready-made test is needed. VCI is the version for claims that a problem is being solved. Case data in Section 8: WHO; CDC *MMWR*; The Carter Center; GPEI and the WHO Polio IHR Emergency Committee; *PLoS Medicine* (Najera et al., 2011); Joint Region Marianas and USGS.

Changelog

Each change names the objection that caused it. A framework that gets stronger when it is hit is a progressive programme in Lakatos's sense.

1. Verification and validation now cite IEEE Std 1012 for both. ASME V&V 10 and 20 are domain standards (solid mechanics; fluids and heat transfer), each covering both.
2. Control test cites the ABAB reversal design and Mill's method of difference.
3. Maintenance rule added so hardware with scheduled upkeep is not misread as preservation.
4. K1 assigns the lane and checks the label; it no longer kills Adaptive problems outright.
5. Claim types and the Mislabeled verdict added after the relabeling dodge was found.
6. Research lane added, with perpetual study, file drawer, and permanent pilot rules.
7. Escape hatches limited to before commitment.
8. Gate 7 measured by candidates eliminated; two empty phases in a row made fatal.
9. Decomposition, proxy owner, evaluator stake, and "Terminal is not cheap" added after the four-problem stress test (brown tree snake, cyber attacks, digital privacy, AI).
10. Intervention-created vectors must be counted, after the polio cVDPV case.
11. Owner defined as whoever bears the cost, after the funder-as-preserver case.
12. Verdict language set to structure, not intent.
13. **RC2**. Vector defined, with six roles and the closure test, after the brown tree snake case: its entry routes had not been counted.
14. **RC2**. Inflow lane added for a permanent source outside the owner's reach; management at the doors only (M3).
15. **RC2**. Neglect ruled out as a vector or lane.
16. **RC2**. Gates 5 and 6 may be marked as not applying only when no field results exist; bench results belong to Gate 4.
17. **RC3**. Licensing split set: method open, product layer closed, names reserved.

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